

Macroecology analysis coursework.pdf

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Macroecology Analysis on Trochilidae Family

title is nice and snappy but a bit vague and doesn't sound like proper English

INTRODUCTION

I explained that no scientific writing starts with the study system!

Macroecology explores large-scale patterns in biodiversity, examining how species distribution and traits like body size and range size vary across environmental gradients. The Trochilidae family, or hummingbirds, is one of the most species-rich and arguably the most remarkable examples of evolutionary radiation among birds. Hummingbirds are found from Alaska (61.8° N) to Tierra del Fuego (55.8° S). According to Sibley (1990), 76% of the 319 recognized species are found in South America. This nectarivorous-insectivorous clade (Bleiweiss, Kirsch & Matheus, 1997) presents a suitable opportunity to test macroecological theories due to their relatively well known geographical ranges of species and nearly completed taxonomic inventory. This report aims to test three major macroecological hypotheses—the Latitudinal Diversity Gradient (LDG), Bergmann's Rule, and Rapoport's Rule—to understand how hummingbird's body size, range size, and species richness vary with latitude.

The LDG describes the pattern where species richness peaks near the equator and declines towards the poles. The hypothesis to be tested in this study is that the Trochilidae family will show a negative relationship between species richness and absolute latitude, consistent with the general LDG pattern. Hummingbirds are predominantly found in tropical and subtropical regions. The stable climate and abundant floral resources in these regions may explain their high species richness. If the hypothesis is supported, it would reinforce the idea that tropical regions foster higher biodiversity, offering insights into the underlying ecological factors driving this trend.

Bergmann's Rule predicts that animals living in colder climates tend to have larger body sizes to conserve heat (Bergmann, 1847). While this rule is typically tested in larger-bodied animals, applying it to hummingbirds, known for their small size and high metabolic demands, provides a novel perspective. Given that hummingbirds are energy-intensive, small-bodied animals, we hypothesize that body size in the Trochilidae family increases towards higher latitudes. However, the small size of hummingbirds and their highly specialized ecology could result in deviations from traditional Bergmannian patterns, offering an opportunity to explore whether the rule holds for a small, tropical-dominant family. Understanding these deviations could reveal important ecological and physiological factors that influence hummingbird body size, challenging the universality of Bergmann's Rule.

Rapoport's Rule suggests that species in colder regions tend to have larger range sizes, likely due to harsher climates requiring greater adaptability (Rapoport, 1982). However, since hummingbirds are concentrated in tropical regions with small, specialized ranges, we hypothesize that hummingbirds will not follow Rapoport's Rule, and instead show a negative relationship between range size and latitude. This hypothesis reflects the observation that hummingbirds, with their strong ecological dependence on specific floral resources, may not exhibit the broad range sizes characteristic of animals in colder regions. Testing this hypothesis could offer insights into the unique

don't dive into the study system until 2nd or 3rd para.
First of all outline the broad topic.
why there is a debate.
highlight mixed findings from different groups and controversy about the mechanisms.
plenty of papers to choose from.

ecological and evolutionary drivers shaping range size in tropical species.

This study uses data on hummingbirds to explore how body size, range size, and species richness vary with latitude. By focusing on the Trochilidae family, we aim to assess how well these species conform to the macroecological rules of LDG, Bergmann's Rule, and Rapoport's Rule, while also investigating potential deviations from these patterns. The results could enhance our understanding of the ecological and evolutionary processes driving hummingbird diversity and distribution, shedding light on the complexities of macroecological theories when applied to highly specialized taxa like hummingbirds.

good intro but over the word limits suggested

METHODS

This study examined the relationships between species richness, body mass, range size, and absolute latitude for the Trochilidae family, testing the LDG, Bergmann's Rule, and Rapoport's Rule. Data preparation, analysis, and visualisation were conducted in R using multiple libraries, including dplyr for data manipulation, ggplot2 for visualisation, caper for phylogenetic analyses, and fasterize for spatial rasterisation. The AVONET dataset was filtered for Trochilidae species (Tobias et al., 2022), and the all-bird phylogeny was used for Phylogenetic Generalized Least Squares (PGLS) regression (Jetz et al., 2012). Absolute latitude was calculated based on the centroid latitude of each species.

good

Species Richness Analysis

For the species richness analysis, latitude was divided into 5° bins to group species by their distance from the equator, with bins labelled by their upper limits. Species richness was calculated by counting the number of species in each bin. A generalised linear model (GLM) with a Poisson distribution was used to examine the relationship between latitude and species richness. Predictions from the model were visualised with trends and standard errors shown as shaded areas around the fitted values. A scatter plot was generated using ggplot2 to illustrate the richness trends, with confidence intervals displayed. Since species richness within each latitudinal bin is independent of phylogeny, no phylogenetic relationship was considered. Additionally, spatial distribution maps of species richness were created, using rasterised data with a colour gradient to represent low to high richness.

clearly explained in simple language - good

Testing Bergmann's and Rapoport's Rules with PGLS

Testing Bergmann's and Rapoport's rules followed similar methods, with body mass and range size as the dependent variables, respectively. Body mass of Trochilidae species was used to test Bergmann's rule. Both body mass and range size data were log-transformed for normalisation. Linear

regression was applied to assess the relationship between absolute latitude and range size. Given the importance of phylogeny, lambda profiling was conducted to optimise the phylogenetic signal, and PGLS regression was used to test both hypotheses while accounting for evolutionary relationships. Finally, linear and phylogenetic fits were compared through visualisation.

RESULTS

Latitudinal Diversity Gradient

always convert p values from these ridiculous numbers to standard notation, keep it simple (eg $p < 0.001$ or whatever)

The GLM model showed a significant negative relationship between species richness and latitude, with an estimate of -0.9004 and a p-value of less than $2e^{-16}$. This means that the closer to the equator, the number of hummingbird species increases, which is supported by a scatter plot and a species richness map (Fig. 1 & 2). At 4° latitude, species diversity reaches its peak.

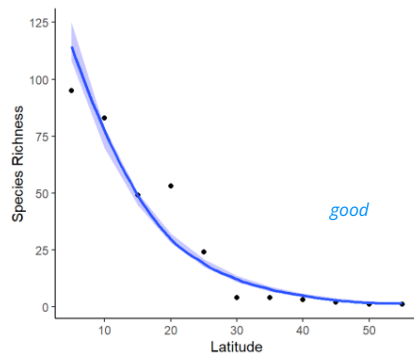


Fig. 1 Relationship between the species richness of Trochilidae and absolute latitude. The graph illustrates how hummingbird species richness is distributed across 5° latitude bins. A negative correlation between species richness and absolute latitude is observed.

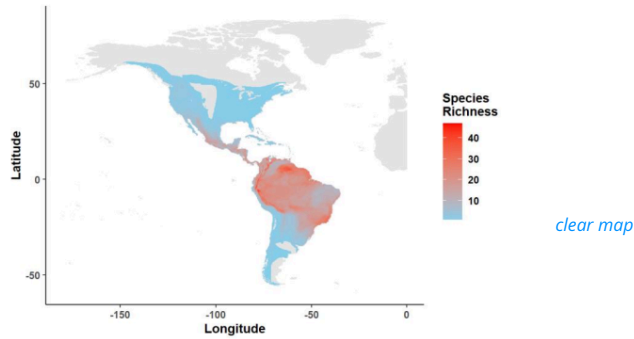


Fig. 2 Species richness hotspot map of Trochilidae. This map illustrates the geographical distribution of Trochilidae species richness. Regions with higher species richness are indicated by more intense red colouring, while areas with lower richness are shaded in blue.

Bergmann's Rule

The LM showed a significant negative relationship between body mass and latitude (estimate = -0.007077 , $p = 0.00763$), suggesting that larger-bodied hummingbirds live in colder areas. However, when the PGLS model accounted for evolutionary relationships, the correlation became non-significant (estimate = 0.0015301 , $p = 0.427$; Fig. 3). The lambda value of 0.85 from the PGLS model suggests that body mass is strongly influenced by evolutionary history rather than latitude. The species with the largest body mass has a centroid latitude of -21.7 , indicating its occurrence in the Southern Hemisphere, near the Tropic of Capricorn.

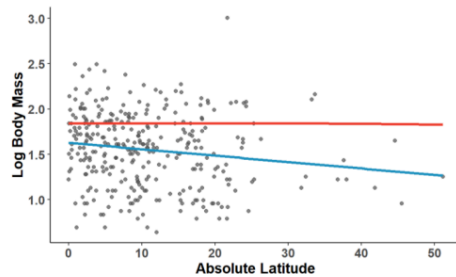


Fig. 3 The relationship between latitude and log body mass of hummingbirds. The graph shows the trends of body mass across latitude using two different models. The blue line represents the weak negative relationship between log body mass and absolute latitude observed with the LM, while the red line illustrates the nearly flat correlation when the PGLS model is used, which considers the evolutionary relationships between species.

Rapoport's Rule

The LM showed no significant relationship between range size and latitude (estimate = 0.01351, $p = 0.4$). The PGLS model also showed no significant result (estimate = 0.014561, $p = 0.385$; Fig. 4). The lambda value of 0.22 suggests that evolutionary relationships do not strongly affect range size, and latitude does not seem to determine the range size for hummingbirds.

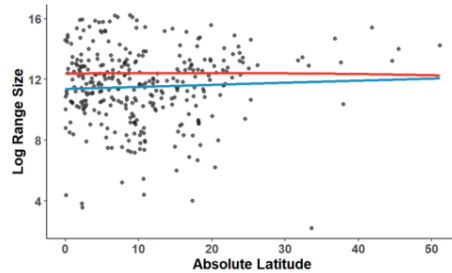


Fig. 4 The relationship between latitude and log(range size). The graph demonstrates the trends between range size and absolute latitude where two different lines indicates different model used. The blue line shows a weak positive correlation between range size and absolute latitude using LM whereas when considering the phylogenetic relationship, the red line shows much flatter.

Table 1 Statistical Models and Results for Hypothesis Testing of Macroecological Rules. This table summarises the statistical estimates and significance levels for three macroecological hypotheses: the LDG, Bergmann's Rule, and Rapoport's Rule. Various models were used, including Generalised Linear Models (GLM), Linear Models (LM), and Phylogenetic Generalised Least Squares (PGLS). The table includes the model estimates, p-values, and their corresponding significance codes: *** ($p < 0.001$), ** ($p < 0.01$), * ($p < 0.05$), . ($p < 0.1$), and no mark ($p \geq 0.1$).

Hypothesis tested	LDG	Bergmann's rule		Rapoport's rule	
Model used	GLM	LM	PGLS	LM	PGLS
Estimate	-0.9004	-0.007077	0.0015301	0.01351	0.014561
p-value	<2e-16 ***	0.00763 **	0.427	0.4	0.385
Signif. codes:	0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

DISCUSSION

well organised results and presentation

Latitudinal Diversity Gradient

The significant negative relationship between species richness and latitude supports the LDG. This pattern is consistent with the idea that tropical regions near the equator provide favourable conditions,

such as stable climates, abundant resources, and higher primary productivity. For hummingbirds, tropical areas likely offer diverse habitats and year-round nectar sources, which promote higher species richness, whereas regions further from the equator may have harsher climates and fewer resources, limiting the number of species that can survive.

However, the relationship between species richness and latitude may be influenced by the spatial scale of analysis and the inclusion of other ecological factors, such as topography. The averaging effect of larger geographical scales may obscure finer patterns of species richness, reducing the power to identify the true drivers of biodiversity. A recent study found that species richness of hummingbirds was positively correlated with both latitude and topographical relief when small quadrats were used, but when topography was excluded, the correlation with latitude became insignificant at coarser spatial scales (Rahbek & Graves, 2000). This suggests that topography could be more important than latitude in determining species richness patterns and scale of analysis plays a crucial role in interpreting macroecological patterns. In the case of hummingbirds, the diverse topography of tropical regions, including mountain ranges and valleys, creates microhabitats that support a wider variety of species. Therefore, a more refined approach that incorporates the interactions between multiple factors and varying spatial scales could offer a clearer understanding to latitudinal gradients in hummingbirds.

good but a bit unclear on the mechanisms explained in lectures promoting high speciation in tropical systems (which tend to shrink ranges because allospecies ranges are predestined to be smaller than ancestral ranges)

Bergmann's Rule

Although the LM showed a significant negative relationship between body mass and latitude, the PGLS model revealed no significant relationship. This discrepancy suggests that factors beyond latitude, such as migratory behaviour and dietary requirements, play a role in shaping hummingbird body size. Migration reduces the need for thermal regulation, making body size less critical for maintaining optimal temperatures (Hiebert, 1993). Furthermore, as nectar feeders, hummingbirds benefit from being small, as larger sizes would require substantially more energy, which may not be sustainable given nectar availability and their high-energy demands during migration. Further investigation could focus on other parameters for body size, such as wing length or bill size, to test Bergmann's rule.

These adaptations, including small body size, high energy efficiency, and migratory behaviour, may also help explain the LDG. Hummingbirds' small size and mobility allow them to exploit diverse floral resources across broad geographical ranges, particularly in the tropics, where resource availability is higher and more stable. This ability to thrive in resource-rich tropical environments may contribute to the higher species richness observed at lower latitudes, aligning with the LDG. Conversely, their specialised feeding and energy requirements may limit their ability to adapt to colder, resource-scarce regions, reinforcing the gradient's decline in diversity towards higher latitudes.

Rapoport's Rule

The lack of a significant relationship between range size and latitude suggests that hummingbirds do not follow Rapoport's Rule. This may be due to their ecological specialization (Sonne et al., 2016). Hummingbirds are highly dependent on nectar-producing plants, and their distributions are closely tied to the availability of these resources. In tropical regions, high plant diversity supports many hummingbird species with smaller, overlapping ranges. In contrast, at higher latitudes, fewer plant species and less stable climates could limit both the number of hummingbird species and their range sizes. Additionally, factors like competition and migration patterns might also play a role. For example, migratory hummingbirds at higher latitudes may not establish large ranges because they only spend part of the year in those regions. These ecological constraints likely override any latitude-related patterns in range size.

This study provides valuable insights into the macroecological patterns of the Trochilidae family. The significant negative relationship between species richness and latitude supports the LDG, suggesting that tropical regions foster higher biodiversity. However, the lack of significant patterns for Bergmann's and Rapoport's rules indicates that factors like migration, ecological specialization, and resource availability may override traditional latitudinal trends in hummingbirds. These findings underscore the complexity of applying macroecological rules to specialized taxa and highlight the need for further research using finer scales and consider multiple factors. In conservation, macroecological analyses are crucial for identifying and prioritizing biodiversity hotspots (Brooks et al., 2010), but the scale and quality of data are vital for accurate conclusions. A scientifically based understanding of scaling effects will be essential for improving conservation strategies and addressing the biodiversity crisis.

stretching the word limits, but good text in general

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mostly good depth and formatting of references

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GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

72/100

PAGE 1

Text Comment. add author name

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Text Comment. don't dive into the study system until 2nd or 3rd para. First of all outline the broad topic. why there is a debate. highlight mixed findings from different groups and controversy about the mechanisms. plenty of papers to choose from.

PAGE 2

Text Comment. good intro but over the word limits suggested

Text Comment. good

Text Comment. clearly explained in simple language - good

Text Comment. you can call them hummingbirds, not Trochilidae species

PAGE 3

Text Comment. always convert p values from these ridiculous numbers to standard notation, keep it simple (eg $p < 0.001$ or whatever)

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PAGE 4

Text Comment. clear map

PAGE 5

Text Comment. well organised results and presentation

PAGE 6

Text Comment. good but a bit unclear on the mechanisms explained in lectures promoting high speciation in tropical systems (which tend to shrink ranges because allospecies ranges are predestined to be smaller than ancestral ranges)

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PAGE 7

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PAGE 8

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PAGE 9

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YUKI WU

CRITERIA MET - MARKER 1

How does this work meet the criteria for the grade it has been given?



Excellent report in general. Reasonable presentation of results and figures without doing enough to get a high first.
Clearly written showing good grasp of research methods and writing, enough for a low first.

AREAS TO IMPROVE - MARKER 1

How could this work – and similar work in future – be improved, to gain a higher mark?



Get the structuring right - deal with the broad question first and don't bring in the study system until later. See notes and suggestions on report.

MARK - MARKER 1

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If you agree with the mark awarded by the first marker, indicate "Agree" here and include the mark in the "Out of 100" field above. If you do not agree with the mark awarded by the first marker, indicate "Proposed mark - XX%" here and leave the "Out of 100" field blank. Also include any additional comments for the student in this field.

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